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Identifying and mapping hendra virus strain diversity DOC	1	106 KB

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Background

Hendra virus (HeV) was first described in 1994 following the outbreak of a novel disease fatally affecting horses and humans in south-east Queensland. Sporadic outbreaks continue to be identified (in 1999, 2004, 2006 & 2007), with a total of 30 equine known cases (75% CFR) and 4 known human cases (50% CFR) to date.

Bats (flying foxes) have been identified as the natural host of the virus.

A closely related virus (Nipah virus - NiV) was responsible for a major outbreak of disease in pigs and humans in Malaysia in 1998-9, and more recently has been identified as the cause of seasonal clusters of human disease (with a 75% case fatality rate) in Bangladesh. The strains of Nipah virus identified in Bangladesh differ genetically from those identified in Malaysia, and feature direct bat-human and human-human transmission. In Australia, the existence of Hendra virus strain variation became evident in a recent equine case which was initially mis-diagnosed by animal health authorities, but subsequently confirmed as a 'variant' strain of HeV.

The 'Henipavirus Research Adoption Forum' held by the Australian Biosecurity CRC in July 2007 brought together over 40 delegates from a range of research end-users. Facilitated discussion sessions reviewed both national and international henipavirus research to date, identified gaps in existing knowledge, and proposed future research priorities. There was a consensus by end-users that more information on the ecology of Hendra virus was needed before changes to policy and/or practice could occur. In identifying the most significant research gaps and the highest priorities for further research, there was broad agreement on the need to better understand the extent of HeV strain variation in bats in the Australian region, and associated variability in transmission efficiency and pathogenicity.

Previous attempts to recover Hendra virus/nucleotide sequence from bats in Australia have had limited success. However, our understanding of HeV infection dynamics in bats, in conjunction with sampling and diagnostic approaches to maximize test sensitivity has significantly advanced in recent years.

In this proposal, the key research question is 'What is the diversity of Hendra viruses occurring in Australia'.

Objectives

- sample populations of bats at multiple times and locations
- screen pooled urine/faecal samples by PCR using a ‘generic’ HeV primer set
- map the diversity of identified strains using phylogenetic analyses
- convey findings to end-users to inform risk management and diagnostic test improvements.

Source: https://www.agriculture.gov.au/biosecurity-trade/policy/emergency/wedpp/identifying_and_mapping_hendra_virus_strain_diversity_stage_1